

ADA PLC Guide V3.0

Consolidated Programming Reference — GX Works2

Every Functional Block, One Register Map, One Document:
ADA 3.0's Pointer-Driven Core plus Heater Staging, T_f , E_{total} /Dynamic Power,
and the Auxiliary Stage Register, Carried Forward and Reconciled

Supersedes ADA PLC Guide V1.1 and the ADA 3.0-Only Draft (V2.0)
for the Nebico Oil Melting/Retention System

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This is the single document a programmer needs while implementing the Nebico ADA control system. It merges ADA PLC Guide V1.1's full block set — Heater Staging with its dual anti-chatter latches, T_f thermal blending, E_{total} /Dynamic Power Evaluation, the Auxiliary Stage register, Tank Scoring/Ranking, Height-to-Mass conversion, Solar/Auxiliary Gating, and Fault Detection — with ADA 3.0's pointer-driven, decimal (float) simplification of Sensor Acquisition/Averaging/Rate-Error, Dynamic Water Target, PO Designation/Timer, the sentinel-based Need Scheduler, and the Modbus-node Oil Scheduler. Every register collision between the two source documents is resolved explicitly, not just flagged — see Section 1.4 for the reconciliation log.

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Purpose and Scope

Design rationale and derivations live in the parent architecture documents (ADA Architecture 2.0.1 and ADA 3.0) — this is nomenclature, a consolidated tag/ID lookup, every ladder logic block the program needs in one place, and concrete register addressing. Nothing is deferred to another programming document: where ADA 3.0 resolved or simplified a block, that resolution is what appears below; where ADA 3.0 was silent (Heater Staging, T_f , E_{total} /Dynamic Power, Auxiliary Stage), the block is carried forward from ADA PLC Guide V1.1, adapted only where the shared register map required it.

Reconciliation Log — Collisions Resolved in This Revision

Conflict	Resolution
M4 meaning	V1.1 uses <code>M4_ThrottleOrDivert</code> for the T_f interlock output. ADA 3.0 proposed reusing M4 for the aggregate PO-stored flag, reasoning T_f was out of scope. With T_f now back in scope, that reuse is withdrawn: M4 keeps its original T_f-interlock meaning , and the aggregate PO-stored flag is assigned a new address, M5_POStored , instead.
Tank scoring formula	V1.1 Section 3.15 computes two score variants per tank (<code>ScoreCasual</code> , weighted 0.6/0.4, and <code>ScoreImm</code> , weighted 0.4/0.6) for the old incumbent/challenger switching rule. The architecture document (ADA 2.0.1 Section 9.1) states explicitly there is only one scoring formula, and ADA 3.0's Need Scheduler confirms the single $0.6 \times \text{Level} + 0.4 \times \text{Temperature}$ form with no variant. The single-formula version is authoritative ; the <code>ScoreImm</code> variant and the old incumbent/challenger threshold rule are dropped, replaced outright by the Need Scheduler's sentinel-based arbiter (Section 3.19).
Dynamic Water Target source	V1.1 Section 3.8 (<code>D_Twater_target</code>) computes the target from <code>D_Ptotal</code> via a linear formula, which does not match the quadratic $T_{\text{target}}(d)$ law both architecture documents specify. The quadratic law is authoritative (Section 3.8 below); the linear <code>D_Ptotal</code> -based rung is dropped.
Rate/fault checking on the 10 sensor channels	V1.1 Section 3.23 computes per-channel heat/cool rate limits (<code>D_MaxHeatRate</code> / <code>D_MaxCoolRate</code>) as part of Fault Detection. ADA 3.0's Sensor Rate Error block (Section 3.7) resolves this with a cleaner two-sided band check on the same 10 channels. ADA 3.0's version is authoritative ; Fault Detection (Section 3.22) keeps only the mass-anomaly and 75 °C hard-limit checks, matching ADA 2.0.1 Section 12.4's "Remaining Fault Detection" scope.
Oil distribution mechanism	V1.1 Section 3.22 drives PBV100–PBV103 directly from <code>M_DemandX/Y/Z</code> . ADA 3.0 Section 9 states the Oil Scheduler redesign replaces this outright . The old X/Y/Z valve-driving rungs and their M-flags are dropped; Y21–Y24 are no longer driven by this PLC program at all — physical dispensing now happens at each node's own skid, coordinated over Modbus (Section 3.21).
Heater output arbitration	V1.1 drives Y7/Y10/Y20 from the Auxiliary Stage register (<code>D_AuxStage</code> , power-threshold based). The architecture document's own Heater Staging design (level-band lookup with dual anti-chatter latches) was never actually ladderised in V1.1. Heater Staging (Section 3.9) is now the sole driver of Y7/Y10/Y20 ; the Auxiliary Stage register is retained (Section 3.18) but is informational only — <code>D_AuxStage</code> no longer drives an output, avoiding a two-master conflict on the same three coils.
Number format	Every real-valued register (temperature, level, flow, mass, energy, power) now lives in the float (E) pool, matching ADA 3.0's decimal convention. This also removes the 16-bit overflow risk that forced DMUL/DADD/DDIV 32-bit arithmetic throughout E_{total} and T_f in V1.1 — those blocks are re-expressed with EMUL/EADD/EDIV below, same formulas, no scaling tricks needed. Counters, indices, and flags stay integer (D/M) as before.

1 Nomenclature

1.1 Instruction Set

Instruction	Meaning
LD / LDI	Load contact (NO/NC) — starts a rung
AND / ANI	Series AND (NO/NC)
OR / ORI	Parallel OR (NO/NC)
OUT	Drive a coil/flag
SET / RST	Latch ON / latch OFF
MOV	Move/copy an integer value
MUL / DIV / ADD / SUB	16-bit integer arithmetic
DMUL / DADD / DDIV	32-bit integer arithmetic (kept only where a block still uses integer counters/indices)
EMOV / EADD / ESUB / EMUL / EDIV	Single-precision real (float) move/arithmetic on E-register pairs — the primary arithmetic family in this guide
INC	Increment a register by 1
CMP, ED= / ED> / ED< / ED≥ / ED≤	Integer compare (CMP); float compare (ED-family) used wherever an E-register is on either side, per the tolerance-band convention in Section 3.7
PLS	One-scan pulse on a rising edge
FROM / TO	Read/write an intelligent module's buffer memory
FOR / NEXT	Repeat a rung block a fixed number of times; loop body steps V/Z itself, no auto-increment
K	Prefix for an integer constant

1.2 Device Prefixes

Prefix	Device type
X	Physical digital input
Y	Physical digital output
M	Internal relay/flag
D	Data register, 16-bit integer (32-bit pair via DMUL/DADD/DDIV where noted)
E	Float/real register, single-precision, 2-word stride — every temperature, level, flow, mass, energy, and power value in this guide
T	Timer
V, Z	Index register — modifies another device's address at runtime, e.g. E30Z means “E30 + current value of Z”
AI-#	Analog input channel (placeholder, pending real tag numbers)

1.3 Process Terms

Term	Meaning
ADA	Adaptive Demand Allocation Architecture
P1/P2/P3	Priority tiers: T5 trim-heat / PO tank maintenance / T10 palm oil melt
LSM/HSM/FSM	Trunk flow states: Low/High/Full Supply Mode
PO tank	Whichever of TA/TB/TC is currently designated
T_f	Dynamic upper thermal boundary (T5 safety cutoff)
E_{total}	Sum of remaining energy demand across currently-served branches
$P_{\text{base}}, E_{\text{req}}$	E_{total} converted to an average power over the planning window, and the throttled power request derived from it
Eco/Efficiency/Standard/Throttle	Drop/Fall/Throttle — Heating-posture modes, driven by T5's level
Rounding latch	Heater Staging's fill-direction anti-chatter buffer (Section 3.9.2)
Comfort latch	Heater Staging's temperature-driven anti-chatter hold-off (Section 3.9.3)
M4 / M5_POStored	See the Reconciliation Log above — M4 is the T_f interlock, M5_POStored is the aggregate PO-designation flag
Sentinel	The scratch-array overwrite value that forces the PO-designated tank's score to lose the Need Scheduler's comparison, regardless of direction (Section 3.19.2)
Node	One of the Oil Scheduler's three dispense points (N1/N2/N3), each with its own local HMI, Start/Stop and Reset buttons, and Modbus link
Line lock	The shared interlock (M.LineLocked) allowing exactly one node to dispense at a time

2 Tag ID Tables

2.1 Digital Inputs

Tag	Correspondence	Notes
X0	E-STOP	NC, hardwired safety assumed upstream
X1	Automation Enable/Disable	
X2	Circulation Pump (manual)	
X3	Oil Distribution Pump (manual)	Shared trunk pump; still commanded by the master while a node dispenses (Section 3.21)
X4	Jacket Pump (manual)	Unused — Y4 runs continuously (Section 3.4)
X5	AODD Pump (manual)	T10 → T5 transfer
X6	Float — HEA, NC	Warning only
X7	Float — SOL, NC	Warning only
X10	Loop Exit	Manual-mode only
X_Confirm	Confirm PO tank designation	Momentary
X_EStopReset	E-STOP reset	Momentary, must follow physical clear
X_AlarmReset	Master alarm acknowledge	Momentary
X_StartStopN1	Oil Scheduler node	Hardwired, one per node; edge interpreted as Start when idle, Stop when dispensing
X_StartStopN2	Start/Stop	
X_StartStopN3		
X_ResetN1	Oil Scheduler node	Hardwired, one per node; cancels only that node's own pending/active request
X_ResetN2	Reset	
X_ResetN3		

2.2 Digital Outputs

Tag	Correspondence	Notes
Y0	E-STOPPED lamp	
Y1	Automation Enabled lamp	
Y2	Circulation Pump — HSM	
Y3	Oil Distribution Pump active	
Y4	Jacket Pump active	Always on with M2.SystemOK
Y5	AODD Pump active	
Y6	Circulation Pump — LSM	
Y7 / Y10 / Y20	Heating Elements 1/2/3, 12 kW each	Driven solely by Heater Staging (Section 3.9), gated on M2.SystemOK
Y11–Y14	SV100–SV103, T5/TA/TB/TC branch valves	Driven from the Need Scheduler’s winning index (Section 3.19.4); normally-open, energise to close
Y15/Y16	SV104/SV105, solar selector	
Y17	Circulation Pump — FSM	
Y25	Solar Cycle Active lamp	

Y21–Y24 (PBV100–PBV103) are no longer driven by this program. The old X/Y/Z oil valve network is superseded by the Oil Scheduler’s Modbus-node model (Reconciliation Log). If those physical valves are still present in the field, confirm they are either permanently open or re-purposed before commissioning — leaving them under no active PLC control is a real physical change from V1.1, not just a ladder change.

2.3 Analog Inputs — Pointer-Driven Acquisition

Loop	Ch.	Source	Register (engineering units)
RTD → Temp	6	2 built-in PT100/analogue + 4-ch RTD/analogue card	E0, E2, E4, E6, E8, E10 — $\theta_S, \theta_{TA}, \theta_{TB}, \theta_{TC}, \theta_{Sol}, \theta_H$
4AD → Level	4	4-channel analogue card (ultrasonic)	E20, E22, E24, E26 — $\alpha_S, \alpha_{TA}, \alpha_{TB}, \alpha_{TC}$, base-referenced mm
1AD → Flow	1	Analogue flow input	E40

Raw AI counts feed the above via FROM, stride-2, D0–D21, 11 channels — unchanged from V1.1.

2.4 Float (E) Registers by Function

Group	Registers
Acquisition	E0–E10 (temp $\times 6$), E20–E26 (level $\times 4$, already height-from-base – see below), E40 (flow)
Level height derivation (commissioning)	E42–E48 (empty-state raw ultrasonic distance $\times 4$, one per tank) – consumed at Acquisition (Section 3.5) to turn the raw top-down distance into the height-from-base that lands in E20–E26; not touched again downstream
Level % display (HMI)	E60–E66 (max-fill height $\times 4$, mm, commissioning placeholder), E70–E76 (% full $\times 4$, HMI-facing, derived every scan)
Height-to-mass constants (commissioning)	E90 (T5 mass-per-mm-height, kg/mm – density $\times$ constant cross-section), E92 (shared TA/TB/TC mass-per-mm-height, same reasoning, one tank geometry)
Averaging (5 s block)	E100–E110 (temp $\times 6$), E120–E126 (level $\times 4$)
Rate-error previous value	E200–E210 (temp $\times 6$), E220–E226 (level $\times 4$)
Rate-error bounds, temperature (hardwired)	E300–E310 (max $\times 6$), E320–E330 (min $\times 6$)
Rate-error bounds, level (HMI-adjustable)	E500–E506 (max $\times 4$), E520–E526 (min $\times 4$)
Dynamic Water Target	E600 (T_{target} out), E610=70.0, E612=40.0, E614=1000.0 mm, E616=1800.0 mm, E618 (re-derived quadratic coefficient)
Heater Staging latches	E630/E631 (3 $\rightarrow$ 2 nominal/latched, mm), E632/E633 (2 $\rightarrow$ 1), E634/E635 (1 $\rightarrow$ 0), E636=65.0, E638=5.0 (comfort offsets)
PO data set	E700 (θ_{tank}), E702 (m_{tank}), E704 (θ_w)
PO Timer	E135 (melting point, unchanged addr. from V1.1), E30 (target temp, unchanged addr.), E140/E142 ($C_{\text{cal,melt}}/C_{\text{cal,liquid}}$), E150 (D_{PO})
Height-to-mass	E752 (T5 mass), E754/756/758 (TA/TB/TC mass) – all direct products of height $\times$ the E90/E92 constants above, no scratch register needed
E_{total}	E900 (running E_α accumulator), E902–E920 (shared scratch, reused sequentially — see Section 3.15), E906/E907/E908 (E_5 sensible/loss/total), E909/E911/E913 (E_C^* sensible/loss/total), E915 (E_{total} final)
T_f	E922 (M_p), E924 (T5 full-tank mass constant, 4871.0 kg), E926 (T_f output)
Dynamic Power	E928 (P_{baseW}), E930 (P_{base}), E932 ($E_{\text{req,kW}}$), E934 (TempDiff scratch), E936 (Planning Window, seconds, HMI-adjustable, retentive)
Need Scheduler	E800/E802/E804 (raw scores TA/TB/TC), E820/E822/E824 (scratch/sentinel array)

2.5 Data (D) Registers by Function

Group	Registers
AI raw counts	D0–D21 (11 channels, stride 2)
PO designation	D50 (pending), D51 (active)
Mode register	D83 (D_Mode, 0–3: Eco/Efficiency/Standard/Full Throttle)
Auxiliary Stage (informational only)	D99 (D_AuxStage, 0–3)
Need Scheduler	D103 (D_NeedCount), D104 (D_ServeCount), D105 (D_WinIndex)
Suspicion	D109 (D_Suspicion)
PO dispense overrun watch (AODD)	D110 (D_DispensedVol), D111 (D_2xNd)
Oil Scheduler — request amount	D160, D161, D162 (per node, retentive)
Oil Scheduler — active node / try count	D163 (D_ActiveNode), D164 (D_TryCount)
Oil Scheduler — node status code	D165, D166, D167 (0=idle, 1=pumping, 2=halted, 3=comms-fault-retry)
Oil Scheduler — failure counters	D170, D171, D172 (retentive, per node)

2.6 Working Flags (M) by Function

Group	Flags
Safety/mode	M_ESTOP, M2_SystemOK, M3_AutoEnabled, M10_Refresh
PO designation	M0–M3 (M0 = none), M5_POSStored (aggregate, see Reconciliation Log)
T_f /P1	M4 (M4_ThrottleOrDivert), M_TfInterlock, M_BatchRelease
P2/Suspicion	M_HMIPrompt, M_Alarm, M_DisableFeed, M_PromptActive, M_Overrun_Pulse, M_DailyReset_Pulse
Sensor Rate Error	M_RateFaultHigh_Arr0–9, M_RateFaultLow_Arr0–9 (Z0-indexed, 10-point contiguous blocks)
Heater Staging	M_HeaterLatch3to2, M_HeaterLatch2to1, M_HeaterLatch1to0 (rounding latch state), M_ComfortHold (comfort latch state)
Need Scheduler	M_D_T5, M_D_TA/TB/TC (demand aggregation), M_P1Demand, M_P2Demand, M_Serve_T5, M_Serve_TA/TB/TC
Dynamic Power fast override	M_FastOverride_T5, M_FastOverride_C, M_FastOverride_Alpha, M_FastOverride
Remaining Fault Detection	M_FaultLevelTA/TB/TC, M_HighLimitTA/TB/TC, M_TAAActiveTransfer (and TB/TC equivalents)
Oil Scheduler	M_LineLocked, M_DispenseActiveN1/N2/N3, M_CommsFaultN1/N2/N3, M_CooldownN1/N2/N3

2.7 Timers

Timer	Purpose
T0	10 s refresh oscillator (self-resetting)
T5	Suspicion 5-min no-response auto-increment
T50	5 s sensor-averaging block window
T130	PO Timer demand watchdog
T140/T141	Node 1: per-try Modbus completion timeout (approx. 5 s) / post-failure cooldown (approx. 1–2 s)
T142/T143	Node 2: same pair
T144/T145	Node 3: same pair

2.8 Index Registers (V, Z)

Register	Purpose
Z0	Ten-channel sensor loop (Acquisition, Averaging, Rate Error), stride 2
Z1	Three-tank loop — reused sequentially by the Need Scheduler scoring loop, the E_{total} per-tank loop, and the Remaining Fault Detection mass-anomaly loop (safe: these run at different points in scan order, never concurrently), stride 2
Z2–Z7, V0–V7	Unused, reserved

3 Ladder Logic — Full Listing

3.1 First-Scan Initialization

```

LD M8002
RST M_Alarm
RST M_HighLimitTA RST M_HighLimitTB RST M_HighLimitTC
RST M_FaultLevelTA RST M_FaultLevelTB RST M_FaultLevelTC
RST M_HeaterLatch3to2 RST M_HeaterLatch2to1 RST M_HeaterLatch1to0
RST M_ComfortHold
MOV K0 D_Suspicion
MOV K0 D_AuxStage
MOV K0 D103 // D_NeedCount
MOV K0 D104 // D_ServeCount
MOV K0 D105 // D_WinIndex
MOV K0 D163 // D_ActiveNode
MOV K0 D164 // D_TryCount

// Hardwired temperature rate bounds -- written once, not operator-facing
EMOV K3.0 E300 // repeat for E302..E310 (max) and E320..E330 (min)
// per commissioning sign-off

// PO Timer reference constants -- unchanged values/addresses from V1.1
EMOV K36.0 E135 // melting point, deg C
EMOV K40.0 E30 // target temperature, deg C

// Planning window default -- first cold-start only, never overwrites an
// operator value on subsequent power-ups (confirm M8002 behaviour on the
// actual Coolmay L01S before relying on this)
EMOV K300.0 E936

```

```
// T5 full-tank mass constant, height-to-mass conversion
EMOV K4871.0 E924
```

3.2 Master Safety and Mode Control

Unchanged from V1.1.

```
LDI X0
SET M_ESTOP
LD X_EStopReset
RST M_ESTOP
LD M_ESTOP
OUT Y0

LDI M_ESTOP
OUT M2_SystemOK

LD X1
OUT Y1
LD X1
OUT M3_AutoEnabled
```

3.3 Manual/Automatic Gating Pattern

Repeat per pump/valve not otherwise driven by a dedicated scheduler block below.

```
LD M2_SystemOK
ANI M3_AutoEnabled
AND X2
OUT [pump output]

LD M2_SystemOK
AND M3_AutoEnabled
AND [arbiter / need-array condition]
OUT [pump output]
```

3.4 Jacket Pump

```
LD M2_SystemOK
OUT Y4
```

3.5 Sensor Data Acquisition

Pointer-driven, ADA 3.0. One loop per physical path.

```
// --- RTD -> Temp, 6 channels ---
FROM K0 K6 D0 K1
MOV K0 Z0
FOR K6
    EMOV DOZ EOZ           // per-sensor scale/offset applied here, not a bare
                           // copy -- see commissioning scaling sheet
    INC Z0
    INC Z0
NEXT

// --- 4AD -> Level, 4 channels ---
```

```

// Raw ultrasonic reading is TOP-DOWN (sensor to liquid surface): maximum
// at empty, minimum at full. E42/44/46/48 hold each tank's empty-state
// distance (commissioning placeholder). Height-from-base = empty distance
// minus raw distance -- this is the only place that subtraction happens;
// E20--E26 downstream are already the resolved height, nothing further
// to undo.
FROM K6 K4 D12 K1
MOV K0 Z0
FOR K4
  EMOV D12Z E980          // raw ultrasonic distance, scratch
  ESUB E42Z E980  E20Z    // height-from-base -> the level register
  INC Z0
  INC Z0
NEXT

// --- 1AD -> Flow, 1 channel ---
FROM K10 K1 D20 K1
EMOV D20 E40

```

3.6 Level Percentage Display (HMI)

Four registers, one per tank, purely for the HMI – independent of the mass calculation in Section 3.14. E60/62/64/66 hold each tank's maximum-fill height (mm from base), a commissioning placeholder; the level register (already height-from-base, Section 3.5) is divided directly against it. Uses the same fresh Z0 (0/2/4/6) indexing as the level acquisition loop above.

```

MOV K0 Z0
FOR K4
  EDIV E20Z E60Z  E982      // height / max-fill height
  EMUL E982 K100.0 E70Z    // -> percent; E70/72/74/76, displayed as a
                          // whole number, held internally as float

  INC Z0
  INC Z0
NEXT

```

3.7 Sensor Averaging

Pointer-driven, ADA 3.0, 5 s block average on the 10 calculation-feeding channels. Flow is excluded.

```

LD  T50
OUT T50 K50          // 5.0 s, self-resetting
LD  T50
MOV K0 Z0
FOR K10
  EADD E100Z E0Z E100Z    // running sum accumulates into the average slot
  INC Z0
  INC Z0
NEXT
// on the 5th accumulation, divide by 5 and reset -- counter register
// omitted here for brevity, same block-average mechanism as ADA 2.0.1 Sec 5

```

3.8 Sensor Rate Error Calculations

Resolved, ADA 3.0. Ten channels, two-sided band, one pointer loop; this block supersedes V1.1's heat/cool rate check inside Fault Detection (Reconciliation Log).

```

MOV K0 Z0

```

```

FOR K10
  ESUB E100Z E200Z E950      // rate this cycle = current avg - prev avg
  MOV  E100Z E200Z          // store current avg as next cycle's "prev"

  LD E950 > E300Z            // rate above max bound (channels 0-5: temp,
  OUT M_RateFaultHigh_ArrZ   // hardwired; channels 6-9: level, HMI --
                              // selector on Z0<K12 needed, see warnbox

  LD E950 < E320Z
  OUT M_RateFaultLow_ArrZ

  INC Z0
  INC Z0
NEXT

```

The loop body must select the correct bound array (E300/E320 hardwired vs. E500/E520 HMI-adjustable) by branching on Z0's current value, since the two groups are not laid out contiguously with each other. Insert that selector before the two compares in the real rung.

3.9 Dynamic Water Target

Quadratic taper law, authoritative version (Reconciliation Log): flat 70.0 °C below 50% fill, quadratic taper to 40.0 °C by 90% fill, flat above — now compared directly against T5's float mm level.

```

LD  E20 < E614              // T5 level < 1000.0 mm
EMOV K70.0 E600

LD  E20 >= E614
AND E20 < E616              // 1000.0 <= level < 1800.0 mm
ESUB E20 E614 E952          // (level - 1000.0)
EMUL E952 E952 E954         // squared
EMUL E954 E618 E956         // x re-derived coefficient
ESUB K70.0 E956 E600

LD  E20 >= E616
EMOV K40.0 E600

```

E618 (the quadratic coefficient) needs re-derivation before commissioning. The original $\frac{3}{2560}$ constant was fit to the old 5 mm-division register $d \in [0, 400]$; comparing directly against raw mm requires refitting so continuity still holds at both knots (70.0 at 1000 mm, 40.0 at 1800 mm).

3.10 Heater Staging: Level-Band Lookup with Dual Latches

Not ladderised in V1.1 — implemented here for the first time, following ADA Architecture 2.0.1 Section 4 directly. Three heater elements staged off T5's level (E20), active band 50%–90% full (1000–1800 mm) split into three equal thirds, mirrored to mm from the original division-based table.

3.10.1 Base Lookup

Level band (mm)	Heaters active
level < 1265	3 (all heaters)
1265 ≤ level < 1530	2
1530 ≤ level < 1800	1
level ≥ 1800	0

3.10.2 Rounding Latch (Anti-Chatter, Filling Direction)

```
// 3 -> 2: nominal 1265 mm, latched release 1300 mm (E630/E631)
LD E20 >= E630
ANI M_HeaterLatch3to2
SET M_HeaterLatch3to2
LD E20 < E630
RST M_HeaterLatch3to2

// Actual heater-count decrease from 3 to 2 held off until E20 >= E631
LD M_HeaterLatch3to2
AND E20 >= E631
// -> proceed to comfort-latch check for this transition (Sec 3.9.3)

// 2 -> 1 (E632 nominal / E633 latched) and 1 -> 0 (E634 nominal / E635
// latched) follow the identical pattern, one M_HeaterLatch flag each
```

Per ADA 2.0.1's own open item: the latch above is specified for the filling (count-decreasing) direction only. The draining direction (count-increasing, e.g. 0→1 at 1800 mm) uses the plain nominal boundary with no buffer. Confirm this asymmetry is still intentional before commissioning.

3.10.3 Comfort Latch (Anti-Chatter, Temperature-Driven)

```
// Hold-off once measured temp (E0) actually permits cutting a heater
LD E600 = K70.0 // Ttarget = 70 (level < 50% full)
RST M_ComfortHold // cut immediately, no further hold

LD E600 >= E636 // 65 <= Ttarget < 70
AND E600 < K70.0
LD E0 < K70.0
SET M_ComfortHold // wait until measured theta_S reaches 70.0
LD E0 >= K70.0
RST M_ComfortHold

LD E600 < E636 // Ttarget < 65
EADD E600 E638 E958 // Ttarget + 5.0
LD E0 < E958
SET M_ComfortHold
LD E0 >= E958
RST M_ComfortHold
```

3.10.4 Combined Output

```
// A transition only takes effect once BOTH the rounding latch has
// released AND the comfort latch is clear
LD E20 < E630
OUT Y7 // heater 1 -- band "3 active" not yet exited
LD E20 < E630
OUT Y10 // heater 2
LD E20 < E630
OUT Y20 // heater 3

LD E20 >= E631
ANI M_ComfortHold
RST Y7 // heater 1 drops first at the 3->2 edge
```

```

LD  E20 >= E633
ANI M_ComfortHold
RST Y10                                // heater 2 drops at the 2->1 edge

LD  E20 >= E635
ANI M_ComfortHold
RST Y20                                // heater 3 drops at the 1->0 edge

// All three gated on system-OK regardless of the above
LD  M2_SystemOK
OR  M_ESTOP
ANI M2_SystemOK
RST Y7   RST Y10   RST Y20

```

3.11 Mode Register

Resolved. Runs against T5's % full (E70, Section 3.6) rather than raw mm – the thresholds below are genuinely %-fill values now, not placeholders standing in for an unresolved unit.

```

LD= D83 K0   AND E70 < K85.0  MOV K1 D83
LD= D83 K1   AND E70 >= K90.0  MOV K0 D83
LD= D83 K1   AND E70 < K65.0  MOV K2 D83
LD= D83 K2   AND E70 >= K70.0  MOV K1 D83
LD= D83 K2   AND E70 < K45.0  MOV K3 D83
LD= D83 K3   AND E70 >= K50.0  MOV K2 D83

```

3.12 Suspicion Score

Unchanged from V1.1.

```

LD  M_Overrun_Pulse
INC  D_Suspicion
LD> D_Suspicion K3
MOV  K3 D_Suspicion

LD= D_Suspicion K1  OUT M_HMIPrompt
LD= D_Suspicion K2  OUT M_Alarm
LD= D_Suspicion K3  OUT M_DisableFeed
LD= D_Suspicion K3  OUT M_Alarm

LD  M_PromptActive
OUT  T5 K3000
LD  T5
INC  D_Suspicion
RST  M_PromptActive

LD  M_DailyReset_Pulse
MOV  K0 D_Suspicion

```

3.13 PO Designation

Merges V1.1's designation mechanism with ADA 3.0's aggregate flag and PO data-set copy.

```

LD  X_Confirm
PLS M100
LD  M100

```

```

MOV D50 D51

LD= D51 K0    OUT M0
LD= D51 K1    OUT M1
LD= D51 K2    OUT M2
LD= D51 K3    OUT M3

// Aggregate PO-stored flag -- M5, not M4 (Reconciliation Log)
LD M1 OR M2 OR M3
OUT M5_POStored

// Copy designated tank's level/temp into the PO data set. Mass is no
// longer re-derived here -- Height-to-Mass Conversion (Section 3.14)
// already computes E754/756/758 for all three tanks every scan, so this
// just picks the right one.
LD= D51 K1  EMOV E2  E700    // theta_TA -> theta_tank
LD= D51 K1  EMOV E754 E702    // m_TA -> m_tank
LD= D51 K2  EMOV E4  E700
LD= D51 K2  EMOV E756 E702
LD= D51 K3  EMOV E6  E700
LD= D51 K3  EMOV E758 E702
EMOV E0  E704                // theta_w -- supplying water = T5's own temp

```

3.14 PO Timer Calculation

Melting/liquid two-phase demand function, watchdog on T130 (unchanged pattern from V1.1). Reference constants E135/E30 keep their values.

```

LD E700 < E135                // theta_tank < melting point
ESUB E135 E700 E960           // (theta_melt - theta_tank)
EMUL E960 E140 E150           // C_cal,melt x demand term A (sketch)

LD E700 >= E135
AND E700 < E30                // melt <= theta_tank < target
ESUB E30 E700 E962
EMUL E962 E142 E150           // C_cal,liquid x demand term B

LD E700 >= E30
EMOV K0.0 E150                // demand satisfied

LD E150 > K0.0
OUT T130 K600
LD T130
RST M_P2Demand
LD X_AlarmReset
RST T130

```

3.15 Height-to-Mass Conversion

Fixed. The V1.1-carried version re-subtracted E20 from a second constant (ESUB K2000.0 E20 E750) before multiplying – but E20 is already height-from-base as of Section 3.5’s fix, so that second subtraction silently corrupted the result. Height-to-mass is now a direct multiply: height $\times$ a per-tank mass-per-mm constant. Diameter is constant with height, so that constant (density $\times$ cross-sectional area) is fixed per tank geometry – but it’s held in a register (E90/E92), not baked into the ladder as a literal K value, so a geometry correction at commissioning doesn’t require a program change.

```

EMUL E20 E90 E752      // T5 mass, kg -- E90: T5 mass-per-mm (commissioning)
EMUL E22 E92 E754      // TA mass, kg -- E92: shared TA/TB/TC mass-per-mm
EMUL E24 E92 E756      // TB mass, kg
EMUL E26 E92 E758      // TC mass, kg

```

E752 (T5) is the one continuously needed downstream (E_{total} , T_f , Section 3.15/3.16). E754/756/758 (TA/TB/TC) are computed every scan for completeness but only the currently-designated tank's value actually gets used, via the PO Designation copy below.

3.16 E_{total} Calculation

Same three-thermal-mass structure as V1.1 Section 3.11, re-expressed in float (no DMUL/DADD/D-DIV scaling needed — see Reconciliation Log). Uses the Z1 three-tank loop.

```

EMOV K0.0 E900
MOV K0 Z1
FOR K3
  LD M_Serve_T_ArrZ1
  ANI M_PODesig_ArrZ1      // M1Z1 -- M1/M2/M3 contiguous
  AND E_ThetaT_ArrZ1 <= K36.0
  ESUB K36.0 E_ThetaT_ArrZ1 E902
  EMUL E902 K2.0 E904      // cp_solid
  EADD E904 K158.4 E906    // + latent/loss allowance
  EMUL E906 E_MassT_ArrZ1 E908
  EADD E900 E908 E900

  LD M_Serve_T_ArrZ1
  ANI M_PODesig_ArrZ1
  AND E_ThetaT_ArrZ1 > K36.0
  AND E_ThetaT_ArrZ1 <= K40.0
  ESUB K40.0 E_ThetaT_ArrZ1 E910
  EMUL E910 K2.1 E908      // cp_liquid
  EMUL E908 E_MassT_ArrZ1 E908
  EADD E900 E908 E900

  INC Z1
  INC Z1
NEXT

// E_5 -- T5 sensible + loss term, window-scaled
ESUB E926 E0 E912          // (Tf - ThetaS)
EMUL E912 K2.1 E906
EMUL E906 E752 E906        // x T5 mass -> E5sensible
ESUB E0 K18.1 E914
EMUL K0.978 E914 E907
EMUL E907 E936 E907        // x planning window
EADD E906 E907 E908        // E5 final

// E_C_star -- PO tank, identical pattern
ESUB K30.0 E700 E916
EMUL E916 K2.1 E909
EMUL E909 E702 E909
ESUB E700 K18.1 E918
EMUL K5.908 E918 E911
EMUL E911 E936 E911
EADD E909 E911 E913        // EC_star final

EADD E900 E908 E915
EADD E915 E913 E915        // E_total final

```

M_PODesig_ArrZ1 is M1/M2/M3, already contiguous, so M1Z1 reaches the right bit for Z1=0/1/2. M_Serve_T_ArrZ1 requires M_Serve_TA/TB/TC to be re-addressed as a contiguous 3-point block for this to work as written — see the Need Scheduler note in Section 3.19.

3.17 T_f Thermal Blending

Unchanged formula from V1.1, re-expressed in float.

```

ESUB E924 E752 E922          // M_p = T5 full mass - T5 current mass
EMUL E922 K2000.0 E962
EMUL E962 E8      E964      // x theta_H (solar loop)
EMUL K75808.0 E0  E966      // x theta_S
EADD E966 E964 E968
EMUL E922 K2000.0 E970
EADD K75808.0 E970 E972
EDIV E968 E972 E926          // Tf final

LD  E926 >= K55.0
OUT M_TfInterlock
LD  M_TfInterlock
OUT M4                      // M4_ThrottleOrDivert -- see Reconciliation Log

```

3.18 Dynamic Power Evaluation

Register-based planning window (E936), unchanged shape from V1.1.

```

LD  M10_Refresh
EDIV E915 E936 E928          // Etotal / window = Watts
EDIV E928 K1000.0 E930        // -> kW

LD  M10_Refresh
AND E81 < K40.0              // measured Twater < 40.0 C
EMOV E930 E932

LD  M10_Refresh
AND E81 >= K70.0             // measured Twater >= 70.0 C
EMOV K0.0 E932

LD  M10_Refresh
AND E81 >= K40.0
AND E81 < K70.0
ESUB K70.0 E81 E934
EMUL E930 E934 E958
EDIV E958 E936 E932

// Fast override, all three duties
LD  M10_Refresh AND M_Serve_T5 AND E0 < E600
OUT M_FastOverride_T5
LD  M10_Refresh AND M_P2Demand AND E700 < K30.0
OUT M_FastOverride_C
// M_FastOverride_Alpha set inside the E_total loop (Sec 3.15); reset here
// before the loop re-evaluates it each scan, placed immediately before
// that loop in the real scan order
LD  M_FastOverride_T5 OR M_FastOverride_C OR M_FastOverride_Alpha
OUT M_FastOverride
LD  M_FastOverride
EMOV K0.0 E932

```

3.19 Auxiliary Stage Register — Informational Only

No longer drives Y7/Y10/Y20 (Reconciliation Log — Heater Staging, Section 3.9, is the sole driver). Retained purely as an HMI-facing power staging indicator derived from E_{req} .

```
LD= D_AuxStage K0 AND E932 > K0.0 MOV K1 D_AuxStage
LD= D_AuxStage K1 AND E932 <= K2.0 MOV K0 D_AuxStage
LD= D_AuxStage K1 AND E932 >= K14.0 MOV K2 D_AuxStage
LD= D_AuxStage K2 AND E932 < K10.0 MOV K1 D_AuxStage
LD= D_AuxStage K2 AND E932 >= K26.0 MOV K3 D_AuxStage
LD= D_AuxStage K3 AND E932 < K22.0 MOV K2 D_AuxStage
```

3.20 Need Scheduler

Sentinel-based single-loop arbiter, ADA 3.0, resolved. Requires M_Serve_TA/TB/TC and the level/temp channels addressed as contiguous 3-point blocks for Z1-indexing to reach them directly.

3.20.1 Scoring

```
MOV K0 Z1
FOR K3
  EMUL E22Z1 K0.6 E940          // 0.6 x level (level channels start at E22)
  EMUL E22Z1 K0.4 E942          // 0.4 x temp (temp channels start at E2)
  EADD E940 E942 E800Z1         // raw score for this tank
  INC Z1
  INC Z1
NEXT
```

3.20.2 Sentinel Exclusion

```
MOV K0 Z1
FOR K3
  EMOV E800Z1 E820Z1
  INC Z1
  INC Z1
NEXT

LD M5_POStored AND M1 EMOV K99999.0 E820
LD M5_POStored AND M2 EMOV K99999.0 E822
LD M5_POStored AND M3 EMOV K99999.0 E824
```

3.20.3 One Loop, Direction Gated by M5_POStored

```
MOV K1 D105          // D_WinIndex, default TA
EMOV E820 E950
MOV K0 Z1
FOR K3
  LDI M5_POStored          // inactive -> track MAXIMUM
  AND E820Z1 > E950
  EMOV E820Z1 E950
  // + index bookkeeping into D105, see warnbox below

  LD M5_POStored          // active -> track MINIMUM
  AND E820Z1 < E950
```

```

EMOV E820Z1 E950
// + index bookkeeping into D105

INC Z1
INC Z1
NEXT

```

Index bookkeeping ($Z1/2 + 1 \rightarrow D105$) needs a DIV/ADD pair guarded by the same rung condition, executed only when the running best actually changes. Compares are strict ($>/<$), never $\geq/\leq$ — on an equal score, the first tank encountered (lowest Z1) keeps its win, no dedicated tie-break logic.

3.20.4 Mode Count and Output

```

MOV K0 D103
LD M_P1Demand INC D103
LD M_Serve_TA OR M_Serve_TB OR M_Serve_TC INC D103

LD M2_SystemOK AND M3_AutoEnabled AND= D103 K1 OUT Y6
LD M2_SystemOK AND M3_AutoEnabled AND= D103 K2 OUT Y2
LD M2_SystemOK AND M3_AutoEnabled AND= D103 K3 OUT Y17

// Branch valve routing from the winning index -- SV100-103 normally-open,
// energise to CLOSE the unserved path (unchanged polarity from V1.1)
LD<> D105 K1 OUT Y12
LD<> D105 K2 OUT Y13
LD<> D105 K3 OUT Y14

```

3.21 Solar and Auxiliary Gating

Unchanged from V1.1; depends on the Dynamic Water Target output (E600) as before.

```

LD M3_AutoEnabled
AND E8 > E600 // theta_Sol > Ttarget
OUT Y15
LDI M3_AutoEnabled
AND X10
OUT Y15

LD M3_AutoEnabled
AND E8 <= E600
OUT Y16
LDI M3_AutoEnabled
ANI X10
OUT Y16

LD Y15
OUT Y25

```

3.22 Oil Scheduler

Full redesign, ADA 3.0, replacing the old X/Y/Z valve-driven oil distribution outright (Reconciliation Log).

3.22.1 Request Flow and Line Occupancy

```
// Node 1 shown; repeat identically for N2/N3
LD  X_StartStopN1
PLS  M200
LD  M200
ANI  M_LineLocked
MOV  K1 D163                      // D_ActiveNode = 1 (N1)
SET  M_LineLocked
SET  M_DispenseActiveN1
MOV  K0 D164
MOV  K1 D165                      // D_NodeStatusN1 = pumping

LD  M200
AND  M_LineLocked
ANI  M_DispenseActiveN1
MOV  K2 D166                      // other node(s): D_NodeStatus = halted
// presses during "halted" are not evaluated further
```

3.22.2 Modbus Poll, Button-Triggered, up to 5 Tries

Same ADPRW / M8029 completion mechanism the legacy periodic poll used – station select, address, count, buffer – just re-triggered differently. The legacy design fired on a free-running timer chain (rising edge of a 0.1s pulse → SET a flag → 1s delay → ADPRW, retried when a completion timeout reset that flag). This keeps the same arm/fire/complete/timeout shape, just armed by the Start button's rising edge instead of a periodic timer.

```
// M200 (Start button rising edge, Sec 3.21.1) arms try 1; a timed-out
// try re-arms the next one below, up to 5.
LD  M200
SET  M_PollArmedN1

LD  M_PollArmedN1
PLS  M210N1                      // fire once per arm, not every scan
LD  M210N1
INC  D164                      // D_TryCount
ADPRW K1 K3 K280 K10 D200      // Node 1's station/address/count --
                                // confirm against the real Modbus map,
                                // one parameter set per node
OUT  T140 K50                  // per-try completion timeout, ~5 s

LD  M8029                      // instruction complete -- success
MOV  K0 D160                   // D_ReqAmountN1 acknowledged, same step
                                // the poll completes, per your spec

MOV  K0 D164
RST  M_PollArmedN1
RST  T140
```

3.22.3 5-Try Failure Handling

```
LD  T140                      // this try timed out with no M8029
PLS  M211N1                   // one-shot on the timeout edge
LD  M211N1
RST  T140

LD  M211N1
AND  D164 < K5
```

```

SET M_PollArmedN1                // re-arm -> next try fires on next PLS

LD  M211N1
AND D164 >= K5
RST M_LineLocked
RST M_PollArmedN1
MOV K3 D165                      // comms-fault-retry
// do NOT clear D_ReqAmountN1 -- Section 3.21 discussion
MOV K0 D164
SET M_CooldownN1
OUT T141 K15                     // cooldown before Start is accepted again
LD  T141
RST M_CooldownN1
RST T141
INC D170                         // D_FailCountN1, persistent

```

The ADPRW parameters shown (K1 K3 K280 K10) are copied from the legacy periodic poll for illustration only – each node needs its own station number and buffer address confirmed against the real Modbus map. N2/N3 repeat this pair of subsections with their own M_PollArmed/M210/M211 flags and their own timer pair (T142/T143, T144/T145 – see the updated Timers table).

3.22.4 Reset

```

LD  X_ResetN1
RST M_DispenseActiveN1
MOV K0 D160
MOV K0 D165
LD  X_ResetN1
AND D163 = K1
RST M_LineLocked

```

3.23 Remaining Fault Detection

Mass-anomaly and 75°C hard-limit checks only — the heat/cool rate checks previously here are superseded by Sensor Rate Error (Section 3.7; Reconciliation Log).

```

LD  M10_Refresh
ESUB E752 E_MassTA_Prev E980
MOV  E752 E_MassTA_Prev

LD  M_ServeTA_AODD
OR   M_ServeN1 OR M_ServeN2 OR M_ServeN3
OUT  M_TAActiveTransfer

LDI  M_TAActiveTransfer
AND  E980 <> K0.0
OUT  M_FaultLevelTA

LD  E2 > K75.0
OUT  M_HighLimitTA
// (repeat identically for TB, TC)

LD  M_HighLimitTA OR M_HighLimitTB OR M_HighLimitTC
SET  M_Alarm
LD  M_Alarm
RST  Y7  RST Y10  RST Y20

```

3.24 Alarm Acknowledge and Reset

```
LD X_AlarmReset
ANI M_HighLimitTA
ANI M_HighLimitTB
ANI M_HighLimitTC
RST M_Alarm
```

4 Structured Text — Experimental

Unverified. GX Works2's ST support is primarily associated with Mitsubishi's iQ-R/iQ-F family, not the classic FX-series line the ladder above assumes — availability on the Coolmay L01S has not been established. Coverage below is partial by design: the heaviest-arithmetic blocks only.

```
(* Tf Thermal Blending *)
Mp := 4871.0 - MassT5;
Tf := (75808.0*ThetaH + Mp*2000.0*ThetaS) / (75808.0 + Mp*2000.0);
TfInterlock := (Tf >= 55.0);
M4 := TfInterlock;

(* Dynamic Water Target *)
IF LevelT5 < 1000.0 THEN
    Ttarget := 70.0;
ELSIF LevelT5 < 1800.0 THEN
    Ttarget := 70.0 - Q_COEFF * (LevelT5 - 1000.0)**2; (* Q_COEFF: TBD, Sec 3.8 *)
ELSE
    Ttarget := 40.0;
END_IF;

(* Need Scheduler -- sentinel + single loop, sketch *)
FOR i := 0 TO 2 DO
    Score[i] := 0.6*Level[i] + 0.4*Temp[i];
    Scratch[i] := Score[i];
END_FOR;
IF POSTored THEN
    Scratch[DesignatedIndex] := 99999.0;
END_IF;
BestIdx := 0; Best := Scratch[0];
FOR i := 1 TO 2 DO
    IF (NOT POSTored AND Scratch[i] > Best) OR
        (POSTored AND Scratch[i] < Best) THEN
        Best := Scratch[i];
        BestIdx := i;
    END_IF;
END_FOR;
```

5 Register Budget and Consolidated Address Map

Pool	Usage
D0–D127 (general use)	AI raw counts, PO designation, Mode register, Aux Stage (informational), Need Scheduler counters/index, Suspicion, PO dispense-overflow watch, Oil Scheduler active-node/try-count/status — all recalculated every scan or every refresh, correctly placed per the EEPROM write-cycle rule established in V1.1
D128+ (EEP-ROM/retentive)	Oil Scheduler request amounts and failure counters — written only on the event that changes them
E-register pool	Every block in Section 3 above — confirm total E-pool availability against the Coolmay L01S memory map before committing; this guide does not assume unlimited headroom

Open items carried into commissioning:

- Dynamic Water Target’s quadratic coefficient (E618) needs re-derivation for the mm-domain comparison.
- M_Serve_TA/TB/TC and the temp/level/mass channel registers need confirmation as genuinely contiguous 3-point blocks for Z1-indexing to work as written in Sections 3.15 and 3.19.
- The ADPRW station/address/count parameters in the Modbus Poll (Section 3.21.2) are placeholders copied from the legacy periodic poll – each node needs its own values confirmed against the real Modbus map, and the settle/timeout/cooldown timer values (T140–T145) tuned against actual RS485 round-trip time.
- Confirm the E-register pool’s actual size on the target hardware.
- **New this pass:** the empty-state distance constants (E42–E48) and max-fill-height constants (E60–E66) are placeholders – both need real site measurement per tank at commissioning, same as the mass-per-mm constants (E90/E92), which additionally need confirmation against the tanks’ actual diameter and the product density.

Conclusion

1. This is now the complete, single programming reference: every block from ADA PLC Guide V1.1 (Heater Staging with dual latches, T_f , E_{total} / Dynamic Power, Auxiliary Stage, Tank Scoring, Height-to-Mass, Solar/Auxiliary Gating, Fault Detection) sits alongside every block ADA 3.0 resolved or simplified (pointer-driven Acquisition/Averaging/Rate-Error, Dynamic Water Target, PO Designation/Timer, the sentinel-based Need Scheduler, the Modbus-node Oil Scheduler).
2. Every collision between the two source documents is resolved explicitly in the Reconciliation Log, not left ambiguous: M4 keeps its T_f meaning; the aggregate PO flag moves to M5_POStored; the single scoring formula and the quadratic Dynamic Water Target law are authoritative over their superseded V1.1 variants; Sensor Rate Error supersedes the old per-tank heat/cool rate check; and the Oil Scheduler fully replaces the old X/Y/Z valve network.
3. All real-valued arithmetic (temperature, level, flow, mass, energy, power) now runs in the float (E) register family, removing the 16-bit overflow risk that forced 32-bit DMUL/DADD/DDIV scaling throughout E_{total} and T_f in V1.1 — same formulas, no scaling tricks.
4. Heater Staging’s level-band lookup with dual anti-chatter latches is ladderised here for the first time and is now the sole driver of Y7/Y10/Y20; the Auxiliary Stage register survives as an informational HMI value only.

5. Five items remain open for commissioning sign-off (Section 5) — none block continued programming, all must close before this guide is trusted on running hardware.

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Supersedes ADA PLC Guide V1.1 and the ADA 3.0-only draft (V2.0) — this is the sole programming reference for the Nebico ADA control system